Mohamed Brahimi

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Education

University of Toronto Sept. 2025 – Expected May 2029

BSc. Statistics Co-op, Minors in Computer Science, Geographic Information Systems *Toronto ON, CA*

- Relevant Courses: Intro to CS II (data structures, graph algorithms in C; OOP in C++), GIS Principles and Applied Mapping (mapping, spatial analysis, PostgreSQL), Intro to Probability, Calculus II

Experience

Education/Web Coordinator Oct. 2025 – Present

UTSC Sports Business Association *Toronto, ON, CA*

- Built and deployed the club website in Next.js, React, and Tailwind CSS, scoring 95 Performance (desktop) and 100 Accessibility, Best Practices, and SEO on Lighthouse, with image optimization and lazy loading.

Web & IT Developer Jun. 2026 – Aug. 2026

Muslim Welfare Canada *Toronto, ON, CA*

- Deployed and documented a centralized NAS server utilizing RAID 5 for hardware fault tolerance and data redundancy, implementing access controls to secure sensitive files for 100 users across 3 office locations
- Redesigned multiple website pages, and added a fundraising page which raised close to $50,000 for a charity event
- Collaborated with front-end developers on a React-based website overhaul

Projects

Atlas | *React, TypeScript, Mapbox GL JS, GeoJSON, Vite, CSS* Oct. 2026

- Contributed the interactive map to a Gemini-powered news investigation platform built in 48 hours by a team of 4 at Hack the Valley 11
- Built the core Mapbox GL JS map component in React/TypeScript which renders locations through 7 layers, with point clustering and click-to-expand
- Rendered city/region records as boundaries and fixed a bug by splitting antimeridian bounds into MultiPolygons
- Wired the map to backend endpoints and Gemini-driven actions, flying the camera to locations by validated ID rather than model-generated coordinates

TOQuery | *Python, FastAPI, PostgreSQL, PostGIS, Docker* Oct. 2026 - Present

- Building a FastAPI + PostGIS REST API serving Toronto data as GeoJSON
- Wrote a GeoPandas ingestion pipeline that deduplicates records, repairs invalid geometries, validates the CRS, and batch-loads into PostGIS with row-count verification
- Containerizing the API and database with Docker Compose and built a Leaflet demo map that renders layers live

FBRef Stats | *Python, Requests, Pandas, Beautiful Soup 4* Jan. 2025

- Created and published a Python library on PyPI for scraping FBRef soccer data, with 16,500+ downloads
- Built a 12-method API across 10 leagues returning league, squad, and scouting data as Pandas DataFrames
- Parsed leaderboard tables embedded in commented-out HTML, added request throttling to keep scraping reliable, and documented with a README and usage examples

GPS Bike Tracker | *Linux, Python, GeoPandas, Supabase, React, Next.js, Mapbox GL JS* Aug. 2026 - Present

- Built a Raspberry Pi GPS tracker in Python logging real-time location, speed, distance, and trip data
- Designed an offline-first sync pipeline: rides are logged to SQLite with a synced flag and uploaded to Supabase in batches, so interrupted syncs resume where they stopped
- Developing a React/Next.js dashboard that renders synced routes in Mapbox GL JS and computes trip statistics (distance, average/max speed, elevation)

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, HTML/CSS, C & C++ (coursework)
Frameworks & Libraries: React, Next.js, FastAPI, Tailwind CSS, Mapbox GL JS, Pandas, GeoPandas, Requests, BS4
Databases: PostgreSQL, PostGIS, SQLite, Supabase
Dev & GIS Tools: Git, Docker, Linux, Raspberry Pi, QGIS, ArcGIS Pro